United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam Trimester: Fall 2025

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (6 Marks)

Transfer of bus control from processor to device takes **600 ns**. Transfer of bus control from device to processor takes **500 ns**.

a.	Suppose, data is transferred in an alternating pattern of **three bytes and five bytes**. This means that **three bytes** are transferred in the first cycle, then **five bytes** are transferred in the next cycle, followed again by **three bytes**, and so on. The I/O device has a data transfer rate of **40 KB/s**. Calculate the **total time** required to transfer **600 bytes** using **cycle stealing mode.** **[Hint: Try to find the number of cycles required to transfer total data using the pattern given in the question]**	[4]
b.	The **808x microprocessor** reserves **three address pins** for special purposes when using memory-mapped I/O. If the **address bus is 8 bits**, calculate **how much memory is reduced** and **what percentage of the total memory** this reduction represents.	[1+1]

Question 2: Answer all the questions. (6 Marks)

a.	**AX = ABCD H, BX = CFB1 H, CX = E412 H,** Execute each of the following instructions **sequentially,** and determine the flags specified in each part. **Provide a brief explanation for your answer**. **i. ADD BX, AX ; Determine AF and CF.** **ii. SUB CX, BX ; Determine OF and PF.**	[4]
b.	Is it possible for a memory location to have **three different logical addresses** while mapping to a **single physical address**? **Justify your answer.**	[2]

Question 3: Answer all the questions. (6 Marks)

a.	**CS = 35E9 H, DS = 2BC1 H, SS = 1F5B H, ES = 2F1D H** **SP = 3344 H, IP = 2211 H, SI = 11AA H** Find how many physical address slots exist that are **simultaneously overlapped** by all three segments (**common to all three segments**): **the Code Segment, Data Segment and Extra Segment.**	[3]

b.	A **SRAM** has a **data bus of 16 bits** and **64K addressable locations**. A **DRAM** has only **half** as many **data lines** compared to SRAM. DRAM is advertised to have **double** the **memory capacity** compared to SRAM. **Determine** the **width of the address bus** of DRAM and the **total capacity (in MB)** of each memory.	**[3]**

Question 4: Answer all the questions. (6 Marks)

a.	You need to design a system where a microprocessor with a **20-bit address bus** and **32-bit data bus** is interfaced to a **5 KB RAM** system using **linear decoding**, starting from address **00800H**. Each RAM chip has an **8-bit address bus** and a **32-bit data bus**. **Draw a block diagram showing the microprocessor, RAM chips and linear decoding logic**. Ensure the first RAM **starts at 00800H**. **Provide the 4th, 6th addresses of the memory space occupied by the 3rd and last RAM chips.**	**[4+2]**

Question 5: Answer all the questions. (6 Marks)

In Public Safety HQ, an **8086 microprocessor** is used for system control. Denji uses a **Chainsaw Control Sensor** for his chainsaw and Aki uses a **Safety Prediction Sensor** to detect danger early. When an input is detected, sensors send an interrupt signal to the microprocessor. The system supports **only the first 80 interrupt types** and the interrupt vector table starts at physical address **00000H**.

Sensor	Given Information
Denji's sensor	Lower byte of **IP** in **00138H**
Aki's sensor	Upper byte of **CS** in **00193H**

a.	What will be the **last address** of the **Interrupt Vector Table**?	**[2]**
b.	**Find** the **interrupt type number** (in hexadecimal) for Denji's sensor and Aki's sensor. **Check** whether each interrupt is **within the valid range.**	**[2]**
c.	If both sensors generate interrupts at the same time, **state** which interrupt will be **serviced first** and **justify your reasoning. (Assume both are valid interrupts)**	**[2]**